Pavan Siva Kumar *Somana.*

+91 86397 96374
pavansivakumar27@gmail.com
Bhimavaram, Andhra Pradesh, India
LinkedIn · GitHub · Portfolio

AI/ML · COMPUTER VISION · IOT SYSTEMS · SOFTWARE DEVELOPMENT · CSE '26

SUMMARY

Computer Science Engineering student with hands-on experience in computer vision, edge AI deployment, IoT-integrated systems, Flutter-based applications, and software development. Skilled in Python, OpenCV, YOLOv8, ONNX, Firebase, PostgreSQL, REST APIs, and GitHub-based workflows. Published 5 IEEE research papers and seeking remote AI/ML, Python, Computer Vision, IoT, Flutter, or software development internship roles.

EDUCATION

2022–2026 — **B.Tech — Computer Science Engineering**
Kalasalingam Academy of Research and Education
CGPA 7.76

2020–2022 — **Class XII — MPC**
Tirumala Educational Institutions
93.7%

2019–2020 — **10th Standard — SSC**
Bhashyam School, Bhimavaram
100%

SKILLS

Languages	Python, C, C++, Java, Dart, JavaScript
AI/ML	Computer Vision, CNNs, YOLOv8, Object Detection, Image Classification, Edge AI
Tools	PyTorch, Ultralytics YOLO, OpenCV, ONNX, Flutter, Jupyter, Google Colab
IoT	Raspberry Pi, ESP32, Arduino, Sensor Integration, IoT Dashboards
Backend	PostgreSQL, MongoDB, MySQL, Firebase, Supabase, REST APIs
Dev	Git, GitHub, VS Code, Android Studio, Bridge, Blockbench

PROJECTS

01 Green Guardian GO — ML-Powered Reverse Vending Machine

Developed an end-to-end smart recycling system using YOLOv8 on Raspberry Pi for real-time plastic bottle detection and validation. Integrated the ML pipeline with IoT hardware, PostgreSQL backend, and a gamified workflow for session tracking and reward generation.

YOLOv8 · Raspberry Pi · PostgreSQL · Computer Vision · IoT · Edge AI

02 BluNest — Smart Aquarium Care IoT Application

Built a Flutter-based IoT application for aquarium monitoring, sensor data visualization, feeding control, and device status tracking. Integrated Firebase for real-time updates across temperature, turbidity, pH, TDS, and water-level parameters.

Flutter · Firebase · IoT · Sensors · ML Roadmap · IEEE Published

03 Dual-Mode Robotic Arm

Built a robotic arm capable of switching between manual joystick control and AI-assisted pick-and-place operations. Integrated motor control, sensors, and vision-based decision-making for real-time object handling; won 1st Place at the RAS Project Showcase.

Robotics · AI Vision · Motor Control · Sensors · Award Winner

04 Facial Recognition-Based Smart Attendance System

Developed a real-time attendance system using Python, OpenCV, and face_recognition. Automated identity verification and attendance logging with MongoDB integration to reduce manual effort and prevent proxy attendance.

Python · OpenCV · Face Recognition · MongoDB

05 Minecraft Bedrock Addon Development & Game Scripting

Developed custom Minecraft Bedrock addons using JSON-based behavior/resource packs and JavaScript scripting. Designed custom blocks, items, abilities, recipes, textures, and gameplay mechanics while debugging Bedrock API behavior using Bridge and Blockbench.

JavaScript · JSON · Game Scripting · Bridge · Blockbench · API Debugging

ACHIEVEMENTS

01 Finalist, Top 10 of 500+ teams — Ground Reality, Launchpad '25, E-Cell BITS Pilani Hyderabad

02 1st Place — RAS Project Showcase for Dual-Mode Robotic Arm

03 Comfortable with GitHub workflows, cloud tools, technical documentation, debugging, and remote collaboration.

IEEE PUBLICATIONS

01 BluNest: IoT & ML-Based Intelligent Aquarium Management System

02 Fix My Ride: ML-Enabled Platform for Intelligent Automotive Assistance

03 Facial Recognition-Based Smart Attendance System Using MongoDB and OpenCV

04 Dual-Mode Robotic Arm for Manufacturing Industries

05 IoT Enabled Device-to-Device Communication for Smart City Application